which Eq. (12.6a) permits us to write as

$$\mathbf{v}_n(\mathbf{k}) \cdot [\hbar\dot{\mathbf{k}} - e\nabla\phi]. \tag{12.14}$$

This will vanish if

$$\hbar\dot{\mathbf{k}} = e\nabla\phi = -e\mathbf{E}, \tag{12.15}$$

which is Eq. (12.6b) in the absence of a magnetic field. However, (12.15) is not necessary for energy to be conserved, since (12.14) vanishes if any term perpendicular to $\mathbf{v}_n(\mathbf{k})$ is added to (12.15). To justify with rigor that the only additional term should be $[\mathbf{v}_n(\mathbf{k})/c] \times \mathbf{H}$, and that the resulting equation should hold for time-dependent fields as well, is a most difficult matter, which we shall not pursue further. The dissatisfied reader is referred to Appendix H for a further way of rendering the semiclassical equations more plausible. There it is shown that they can be written in a very compact Hamiltonian form. To find a really compelling set of arguments, however, it is necessary to delve rather deeply into the (still growing) literature on the subject.[15]

CONSEQUENCES OF THE SEMICLASSICAL EQUATIONS OF MOTION

The rest of this chapter surveys some of the fundamental direct consequences of the semiclassical equations of motion. In Chapter 13 we shall turn to a more systematic way of extracting theories of conduction.

In most of the discussions that follow we shall consider a single band at a time, and shall therefore drop reference to the band index except when explicitly comparing the properties of two or more bands. For simplicity we shall also take the electronic equilibrium distribution function to be that appropriate to zero temperature. In metals finite temperature effects will have negligible influence on the properties discussed below. Thermoelectric effects in metals will be discussed in Chapter 13, and semiconductors will be treated in Chapter 28.

The spirit of the analysis that follows is quite similar to that in which we discussed transport properties in Chapters 1 and 2: We shall describe collisions in terms of a simple relaxation-time approximation, and focus most of our attention on the motion of electrons between collisions as determined (in contrast to Chapters 1 and 2) by the *semiclassical* equations of motion (12.6).

Filled Bands Are Inert

A filled band is one in which all the energies lie below[16] $\mathcal{E}_F$. Electrons in a filled band with wave vectors in a region of k-space of volume $d\mathbf{k}$ contribute $d\mathbf{k}/4\pi^3$ to the total electronic density (Eq. (12.7)). Thus the number of such electrons in a region of position space of volume $d\mathbf{r}$ will be $d\mathbf{r}\, d\mathbf{k}/4\pi^3$. One can therefore characterize a filled band semiclassically by the fact that the density of electrons in a six-dimensional rk-space (called phase space, in analogy to the rp-space of ordinary classical mechanics) is $1/4\pi^3$.

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[15] See, for example, the references given in footnote 1.

[16] More generally, the energies should be so far below the chemical potential μ compared with $k_B T$ that the Fermi function is indistinguishable from unity throughout the band.

The semiclassical equations (12.6) imply that a filled band remains a filled band at all times, even in the presence of space- and time-dependent electric and magnetic fields. This is a direct consequence of the semiclassical analogue of Liouville's theorem, which asserts the following:[17]

Given any region of six-dimensional phase space Ω_t, consider the point $\mathbf{r}'$, $\mathbf{k}'$ into which each point $\mathbf{r}$, $\mathbf{k}$ in Ω_t is taken by the semiclassical equations of motion between times[18] t and t'. The set of all such points $\mathbf{r}'$, $\mathbf{k}'$ constitutes a new region $\Omega_{t'}$, whose volume is the same as the volume of Ω_t (see Figure 12.2); i.e., phase space volumes are conserved by the semiclassical equations of motion.

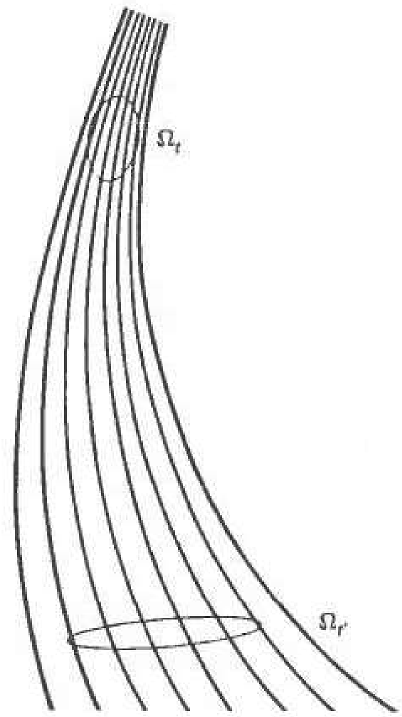

Figure 12.2
Semiclassical trajectories in rk-space. The region $\Omega_{t'}$ contains at time t' just those points that the semiclassical motion has carried from the region Ω_t at time t. Liouville's theorem asserts that Ω_t and $\Omega_{t'}$ have the same volume. (The illustration is for a two-dimensional rk-space lying in the plane of the page, i.e., for semiclassical motion in one dimension.)

This immediately implies that if the phase space density is $1/4\pi^3$ at time zero, it must remain so at all times, for consider any region Ω at time t. The electrons in Ω at time t are just those that were in some other region Ω_0 at time zero where, according to Liouville's theorem, Ω_0 has the same volume as Ω. Since the two regions also have the same number of electrons, they have the same phase space density of electrons. Because that density was $1/4\pi^3$, independent of the region at time 0, it must also be

[17] See Appendix H for a proof that the theorem applies to semiclassical motion. From a quantum mechanical point of view the inertness of filled bands is a simple consequence of the Pauli exclusion principle: The "phase space density" cannot increase if every level contains the maximum number of electrons allowed by the Pauli principle; furthermore, if interband transitions are prohibited, neither can it decrease, for the number of electrons in a level can only be reduced if there are some incompletely filled levels in the band for those electrons to move into. For logical consistency, however, it is necessary to demonstrate that this conclusion also follows directly from the semiclassical equations of motion, without reinvoking the underlying quantum mechanical theory that the model is meant to replace.

[18] The time t' need not be greater than t; i.e., the regions from which Ω_t evolved have the same volume as Ω_t, as well as the regions into which Ω_t will evolve.

$1/4\pi^3$, independent of the region at time t. Thus semiclassical motion between collisions cannot alter the configuration of a filled band, even in the presence of space- and time-dependent external fields.[19]

However, a band with a constant phase space density $1/4\pi^3$ cannot contribute to an electric or thermal current. To see this, note that an infinitesimal phase space volume element $d\mathbf{k}$ about the point $\mathbf{k}$ will contribute $d\mathbf{k}/4\pi^3$ electrons per unit volume, all with velocity $\mathbf{v}(\mathbf{k}) = (1/\hbar)\,\partial\mathcal{E}(\mathbf{k})/\partial\mathbf{k}$ to the current. Summing this over all $\mathbf{k}$ in the Brillouin zone, we find that the total contribution to the electric and energy current densities from a filled band is

$$\mathbf{j} = (-e)\int \frac{d\mathbf{k}}{4\pi^3}\frac{1}{\hbar}\frac{\partial\mathcal{E}}{\partial\mathbf{k}},$$

$$\mathbf{j}_{\mathcal{E}} = \int \frac{d\mathbf{k}}{4\pi^3}\,\mathcal{E}(\mathbf{k})\frac{1}{\hbar}\frac{\partial\mathcal{E}}{\partial\mathbf{k}} = \frac{1}{2}\int \frac{d\mathbf{k}}{4\pi^3}\frac{1}{\hbar}\frac{\partial}{\partial\mathbf{k}}\left(\mathcal{E}(\mathbf{k})\right)^2. \tag{12.16}$$

But both of these vanish as a consequence of the theorem[20] that the integral over any primitive cell of the gradient of a periodic function must vanish.

Thus only partially filled bands need be considered in calculating the electronic properties of a solid. This explains how that mysterious parameter of free electron theory, the number of conduction electrons, is to be arrived at: *Conduction is due only to those electrons that are found in partially filled bands.* The reason Drude's assignment to each atom of a number of conduction electrons equal to its valence is often successful is that in many cases those bands derived from the atomic valence electrons are the only ones that are partially filled.

Evidently a solid in which all bands are completely filled or empty will be an electrical and (at least as far as *electronic* transport of heat is concerned) thermal insulator. Since the number of levels in each band is just twice the number of primitive cells in the crystal, all bands can be filled or empty *only* in solids with an even number of electrons per primitive cell. Note that the converse is not true: Solids with an even number of electrons per primitive cell may be (and frequently are) conductors, since the overlap of band energies can lead to a ground state in which several bands are partially filled (see, for example, Figure 12.3). We have thus derived a necessary, but by no means sufficient, condition for a substance to be an insulator.

It is a reassuring exercise to go through the periodic table looking up the crystal structure of all insulating solid elements. They will all be found to have either even valence or (e.g., the halogens) a crystal structure that can be characterized as a lattice with a basis containing an even number of atoms, thereby confirming this very general rule.

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[19] Collisions cannot alter this stability of filled bands either, provided that we retain our basic assumption (Chapter 1, page 6 and Chapter 13, page 245) that whatever else they do, the collisions cannot alter the distribution of electrons when it has its thermal equilibrium form. For a distribution function with the constant value $1/4\pi^3$ is precisely the zero temperature equilibrium form for any band all of whose energies lie below the Fermi energy.

[20] The theorem is proved in Appendix I. The periodic functions in this case are $\mathcal{E}(\mathbf{k})$ in the case of $\mathbf{j}$, and $\mathcal{E}(\mathbf{k})^2$ in the case of $\mathbf{j}_{\mathcal{E}}$.

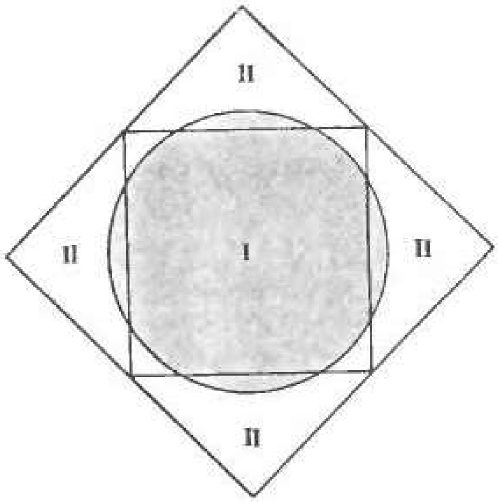

Figure 12.3
A two-dimensional illustration of why a divalent solid can be a conductor. A free electron circle, whose area equals that of the first Brillouin zone (I) of a square Bravais lattice, extends into the second zone (II), thus producing two partially filled bands. Under the influence of a sufficiently strong periodic potential the pockets of first-zone holes and second-zone electrons might shrink to zero. Quite generally, however, a weak periodic potential will always lead to this kind of overlap (except in one dimension).

Semiclassical Motion in an Applied DC Electric Field

In a uniform static electric field the semiclassical equation of motion for **k** (Eq. (12.6)) has the general solution

$$\mathbf{k}(t) = \mathbf{k}(0) - \frac{e\mathbf{E}t}{\hbar}. \tag{12.17}$$

Thus in a time t every electron changes its wave vector by the same amount. This is consistent with our observation that applied fields can have no effect on a filled band in the semiclassical model, for a uniform shift in the wave vector of *every* occupied level does not alter the phase space density of electrons when that density is constant, as it is for a filled band. However, it is somewhat jarring to one's classical intuition that by shifting the wave vector of every electron by the same amount we nevertheless fail to bring about a current-carrying configuration.

To understand this, one must remember that the current carried by an electron is proportional to its velocity, which is not proportional to **k** in the semiclassical model. The velocity of an electron at time t will be

$$\mathbf{v}(\mathbf{k}(t)) = \mathbf{v}\left(\mathbf{k}(0) - \frac{e\mathbf{E}t}{\hbar}\right). \tag{12.18}$$

Since $\mathbf{v}(\mathbf{k})$ is periodic in the reciprocal lattice, the velocity (12.18) is a bounded function of time and, when the field **E** is parallel to a reciprocal lattice vector, oscillatory! This is in striking contrast to the free electron case, where **v** is proportional to **k** and grows linearly in time.

The **k** dependence (and, to within a scale factor, the t dependence) of the velocity is illustrated in Figure 12.4, where both $\mathcal{E}(k)$ and $v(k)$ are plotted in one dimension. Although the velocity is linear in k near the band minimum, it reaches a maximum as the zone boundary is approached, and then drops back down, going to zero at the zone edge. In the region between the maximum of v and the zone edge the velocity actually decreases with increasing k, so that the acceleration of the electron is opposite to the externally applied electric force!

This extraordinary behavior is a consequence of the additional force exerted by the periodic potential, which, though no longer explicit in the semiclassical model, lies buried in it (through the functional form of $\mathcal{E}(\mathbf{k})$). As an electron approaches a

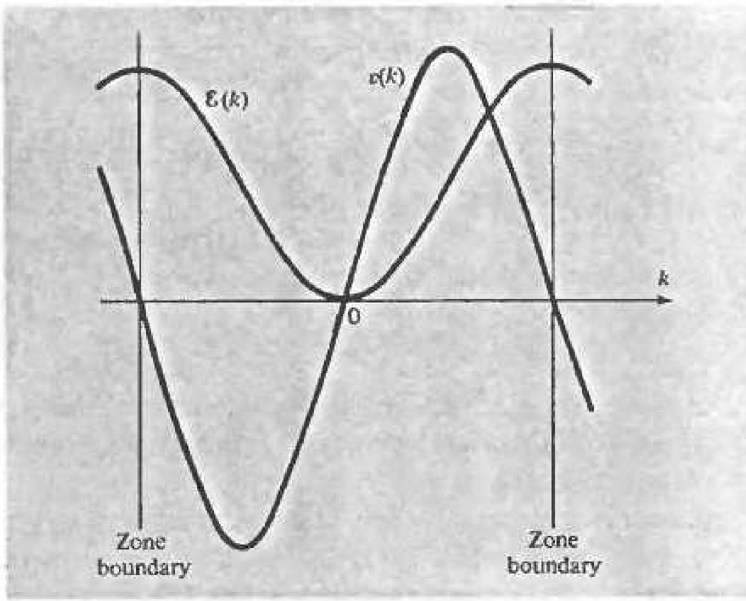

Figure 12.4
$\mathcal{E}(k)$ and $v(k)$ vs. k (or vs. time, via Eq. (12.17)) in one dimension (or three dimensions, in a direction parallel to a reciprocal lattice vector that determines one of the first-zone faces.)

Bragg plane, the external electric field moves it toward levels in which it is increasingly likely to be Bragg-reflected back in the opposite direction.[21]

Thus if an electron could travel between collisions a distance in k-space larger than the dimensions of the zone, it would be possible for a DC field to induce an alternating current. Collisions, however, quite emphatically exclude this possibility. For reasonable values of the field and relaxation time the change in wave vector between two collisions, given by (12.17), is a minute fraction of the zone dimensions.[22]

But although the hypothetical effects of periodic motion in a DC field are inaccessible to observation, effects dominated by electrons that are close enough to the zone boundary to be decelerated by an applied field are readily observable through the curious behavior of "holes."

Holes

One of the most impressive achievements of the semiclassical model is its explanation for phenomena that free electron theory can account for only if the carriers have a positive charge. The most notable of these is the anomalous sign of the Hall coefficient in some metals (see page 58). There are three important points to grasp in understanding how the electrons in a band can contribute to currents in a manner suggestive of positively charged carriers:

1. Since electrons in a volume element $d\mathbf{k}$ about $\mathbf{k}$ contribute $-e\mathbf{v}(\mathbf{k})d\mathbf{k}/4\pi^3$ to the current density, the contribution of all the electrons in a given band to the current density will be

$$\mathbf{j} = (-e)\int_{\text{occupied}} \frac{d\mathbf{k}}{4\pi^3}\,\mathbf{v}(\mathbf{k}), \tag{12.19}$$

[21] For example, it is just in the vicinity of Bragg planes that plane-wave levels with different wave vectors are most strongly mixed in the nearly free electron approximation (Chapter 9).

[22] With an electric field of order 10^{-2} volt/cm and a relaxation time of order 10^{-14} sec, $eE\tau/\hbar$ is of order 10^{-1} cm^{-1}. Zone dimensions are of order $1/a \sim 10^8$ cm^{-1}.

where the integral is over all occupied levels in the band.[23] By exploiting the fact that a completely filled band carries no current,

$$0 = \int_{\text{zone}} \frac{d\mathbf{k}}{4\pi^3}\,\mathbf{v}(\mathbf{k}) = \int_{\text{occupied}} \frac{d\mathbf{k}}{4\pi^3}\,\mathbf{v}(\mathbf{k}) + \int_{\text{unoccupied}} \frac{d\mathbf{k}}{4\pi^3}\,\mathbf{v}(\mathbf{k}), \tag{12.20}$$

we can equally well write (12.19) in the form:

$$\mathbf{j} = (+e) \int_{\text{unoccupied}} \frac{d\mathbf{k}}{4\pi^3}\,\mathbf{v}(\mathbf{k}). \tag{12.21}$$

Thus *the current produced by occupying with electrons a specified set of levels is precisely the same as the current that would be produced if* (*a*) *the specified levels were unoccupied and* (*b*) *all other levels in the band were occupied but with particles of charge* $+e$ *(opposite to the electronic charge).*

Thus, even though the only charge carriers are electrons, we may, whenever it is convenient, consider the current to be carried entirely by fictitious particles of positive charge that fill all those levels in the band that are unoccupied by electrons.[24] The fictitious particles are called *holes.*

When one chooses to regard the current as being carried by positive holes rather than negative electrons, the electrons are best regarded as merely the absence of holes; i.e., levels occupied by electrons are to be considered unoccupied by holes. It must be emphasized that pictures cannot be mixed within a given band. If one wishes to regard electrons as carrying the current, then the unoccupied levels make no contribution; if one wishes to regard the holes as carrying the current, then the electrons make no contribution. One may, however, regard some bands in the electron picture and other bands in the hole picture, as suits one's convenience.

To complete the theory of holes we must consider the manner in which the set of unoccupied levels changes under the influence of applied fields:

2. *The unoccupied levels in a band evolve in time under the influence of applied fields precisely as they would if they were occupied by real electrons* (*of charge* $-e$).

This is because, given the values of $\mathbf{k}$ and $\mathbf{r}$ at $t = 0$, the semiclassical equations, being six first-order equations in six variables, uniquely determine $\mathbf{k}$ and $\mathbf{r}$ at all subsequent (and all prior) times, just as in ordinary classical mechanics the position and momentum of a particle at any instant determine its entire orbit in the presence of specified external fields. In Figure 12.5 we indicate schematically the orbits determined by the semiclassical equations, as lines in a seven-dimensional *rkt*-space. Because any point on an orbit uniquely specifies the entire orbit, two distinct orbits can have no points in common. We can therefore separate the orbits into occupied and unoccupied orbits, according to whether they contain occupied or unoccupied points at $t = 0$. At any time after $t = 0$ the unoccupied levels will lie on unoccupied orbits, and the occupied levels on occupied orbits. Thus the evolution of both occupied and unoccupied levels is completely determined by the structure of the

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[23] This need not be the set of levels with energies less than $\mathcal{E}_F$, since we are interested in the non-equilibrium configurations brought about by applied fields.

[24] Note that this contains as a special case the fact that a filled band can carry no current, for a filled band has no unoccupied levels and therefore no fictitious positive carriers.

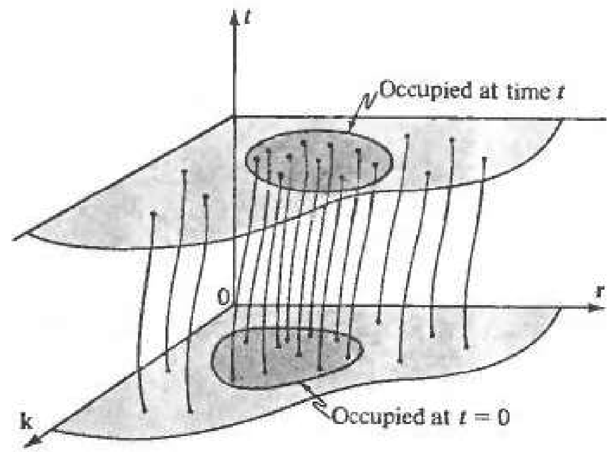

Figure 12.5
Schematic illustration of the time evolution of orbits in semiclassical phase space (here **r** and **k** are each indicated by a single coordinate). The occupied region at time t is determined by the orbits that lie in the occupied region at time $t = 0$.

orbits. But this structure depends only on the form of the semiclassical equations (12.6), and not on whether an electron happens actually to be following a particular orbit.

3. It therefore suffices to examine how electrons respond to applied fields to learn how holes do. The motion of an electron is determined by the semiclassical equation:

$$\hbar \dot{\mathbf{k}} = (-e)\left(\mathbf{E} + \frac{1}{c}\mathbf{v} \times \mathbf{H}\right). \tag{12.22}$$

Whether or not the orbit of the electron resembles that of a free particle of negative charge depends on whether the acceleration, $d\mathbf{v}/dt$, is or is not parallel to $\dot{\mathbf{k}}$. Should the acceleration be opposite to $\dot{\mathbf{k}}$, then the electron would respond more like a positively charged free particle. As it happens, it is often the case that $d\mathbf{v}(\mathbf{k})/dt$ is indeed directed opposite to $\dot{\mathbf{k}}$ when $\mathbf{k}$ is the wave vector of an unoccupied level, for the following reason:

In equilibrium and in configurations that do not deviate substantially from equilibrium (as is generally the case for nonequilibrium electronic configurations of interest) the unoccupied levels often lie near the top of the band. If the band energy $\mathcal{E}(\mathbf{k})$ has its maximum value at $\mathbf{k}_0$, then if $\mathbf{k}$ is sufficiently close to $\mathbf{k}_0$, we may expand $\mathcal{E}(\mathbf{k})$ about $\mathbf{k}_0$. The linear term in $\mathbf{k} - \mathbf{k}_0$ vanishes at a maximum, and if we assume, for the moment, that $\mathbf{k}_0$ is a point of sufficiently high symmetry (e.g., cubic), then the quadratic term will be proportional to $(\mathbf{k} - \mathbf{k}_0)^2$. Thus

$$\mathcal{E}(\mathbf{k}) \approx \mathcal{E}(\mathbf{k}_0) - A(\mathbf{k} - \mathbf{k}_0)^2, \tag{12.23}$$

where A is positive (since $\mathcal{E}$ is maximum at $\mathbf{k}_0$). It is conventional to define a positive quantity m^* with dimensions of mass by:

$$\frac{\hbar^2}{2m^*} = A. \tag{12.24}$$

For levels with wave vectors near $\mathbf{k}_0$,

$$\mathbf{v}(\mathbf{k}) = \frac{1}{\hbar}\frac{\partial \mathcal{E}}{\partial \mathbf{k}} \approx -\frac{\hbar(\mathbf{k} - \mathbf{k}_0)}{m^*}, \tag{12.25}$$

and hence

$$\mathbf{a} = \frac{d}{dt}\mathbf{v}(\mathbf{k}) = -\frac{\hbar}{m^*}\dot{\mathbf{k}}, \tag{12.26}$$

i.e., the acceleration is opposite to $\dot{\mathbf{k}}$.

Substituting the acceleration wave vector relation (12.26) into the equation of motion (12.22) we find that as long as an electron's orbit is confined to levels close enough to the band maximum for the expansion (12.23) to be accurate, the (negatively charged) electron responds to driving fields as if it had a negative mass $-m^*$. By simply changing the sign of both sides, we can equally well (and far more intuitively) regard Eq. (12.22) as describing the motion of a positively charged particle with a positive mass m^*.

Since the response of a hole is the same as the response an electron would have if the electron were in the unoccupied level (point 2 above), this completes the demonstration that holes behave in all respects like ordinary positively charged particles.

The requirement that the unoccupied levels lie sufficiently close to a highly symmetrical band maximum can be relaxed to a considerable degree.[25] We might expect dynamical behavior suggestive of positively or negatively charged particles, according to whether the angle between $\dot{\mathbf{k}}$ and the acceleration is greater than or less than 90° ($\dot{\mathbf{k}} \cdot \mathbf{a}$ negative or positive). Since

$$\dot{\mathbf{k}} \cdot \mathbf{a} = \dot{\mathbf{k}} \cdot \frac{d}{dt}\mathbf{v} = \dot{\mathbf{k}} \cdot \frac{d}{dt}\frac{1}{\hbar}\frac{\partial \mathcal{E}}{\partial \mathbf{k}} = \frac{1}{\hbar}\sum_{ij} \dot{k}_i \frac{\partial^2 \mathcal{E}}{\partial k_i\, \partial k_j}\dot{k}_j, \tag{12.27}$$

a sufficient condition that $\dot{\mathbf{k}} \cdot \mathbf{a}$ be negative is that

$$\sum_{ij} \Delta_i \frac{\partial^2 \mathcal{E}(\mathbf{k})}{\partial k_i\, \partial k_j}\Delta_j < 0 \qquad \text{(for any vector } \Delta). \tag{12.28}$$

When $\mathbf{k}$ is at a local maximum of $\mathcal{E}(\mathbf{k})$, (12.28) must hold, for if the inequality was reversed for any vector Δ_0, then the energy would increase as $\mathbf{k}$ moved from the "maximum" in the direction of Δ_0. By continuity, (12.28) will therefore hold in some neighborhood of the maximum, and we can expect an electron to respond in a manner suggestive of positive charge, provided that its wave vector remains in that neighborhood.

The quantity m^* determining the dynamics of holes near band maxima of high symmetry is known as the "hole effective mass." More generally, one defines an "effective mass tensor":

$$[\mathbf{M}^{-1}(\mathbf{k})]_{ij} = \pm\frac{1}{\hbar^2}\frac{\partial^2 \mathcal{E}(\mathbf{k})}{\partial k_i\, \partial k_j} = \pm\frac{1}{\hbar}\frac{\partial v_i}{\partial k_j}, \tag{12.29}$$

where the sign is $-$ or $+$ according to whether $\mathbf{k}$ is near a band maximum (holes) or minimum (electrons). Since

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \pm\mathbf{M}^{-1}(\mathbf{k})\hbar\dot{\mathbf{k}}, \tag{12.30}$$

[25] If the geometry of the unoccupied region of k-space becomes too complicated, however, the picture of holes becomes of limited usefulness.